October 23, 2026

OSTP Director Kratsios and Science Policy Experts Discuss Institutional Reform, PhD Training, and a New Golden Age of Science.

On Tuesday (October 20), the American Enterprise Institute (AEI) held an event titled “Toward a New Golden Age of Science: A Conversation with Michael Kratsios, Director of the White House Office of Science and Technology Policy.” Kratsios and science policy experts discussed the White House’s Science: A New Golden Age report, including plans to build new research institutions, overhaul PhD training and grant making, and whether these reforms can win durable political support.

Highlights.

- **Fireside Chat with OSTP Director Michael Kratsios.** Kratsios said U.S. science looks very different from the postwar era when the federal government funded about 70% of research and development (R&D). He argued that today’s increasingly multidisciplinary discoveries require new institutions such as X-Labs, which are new labs set up to solve a specific set of problems by bringing together scientist from different fields. He described new PhD pathways that shorten training and place students in industry or national labs, called for more flexible grant durations and faster funding, and pointed to golden tickets that let individual reviewers fund riskier ideas. Kratsios also said the report’s ideas are not partisan and that the National Science Foundation (NSF) is on track to spend the funds Congress appropriated.

- **Public Trust and Stable Funding.** Arizona State University’s Arthur Daemmrich said the Arizona State University Consortium for Science’s public forums found no real rejection of science but that participants want a voice in what research gets funded and support complements to merit review such as golden tickets. The Institute for Progress’ Jenn Gustetic called for an agreed base of stable funding, flexibility for agencies to carry money over between years, and a resource allocation process that runs in public view.

- **Political Durability of Reform.** Vanderbilt’s Asad Ramzanali argued that laws creating institutions and moving billions of dollars are far more durable than executive orders or visions. He said actions such as freezing university funding and calling for deep cuts at NSF undercut the idea of a golden age. He also criticized the firing of the National Science Board (NSB), arguing Congress should change the law instead, while Gustetic said metascience reforms must be sustainable and bipartisan to last.

- **Commercialization and the Science Workforce.** During audience questions, Daemmrich said federal labs and universities have become an intellectual property pricing scheme and called for a return to a model where researchers spend time in both industry and academia. Gustetic asked whether scientists can be trained to be better businesspeople. Panelists also stressed that science and technology careers are not limited to PhDs and that the U.S. undervalues craft, the hands-on skill of building and working with scientific equipment, as a career path.

See below for more details from our select coverage of the event. As always, please reach out with any questions.

Fireside Chat with OSTP Director Michael Kratsios.

- Michael Kratsios, Director, White House Office of Science and Technology Policy (OSTP)
- M. Anthony Mills, Director, Center for Technology, Science, and Energy, American Enterprise Institute (*Moderator*)

Why a New Golden Age Now
Mills asked why Kratsios wrote the report now and how he thinks about a new golden age of science. Kratsios said the most recognizable feature of the old golden age was that the federal government funded about 70% of R&D because of the war effort. He said the way science is approached today is very different from the 1950s and 1960s. He noted that the private sector now funds the majority of R&D and said today's era differs in two ways. The first is where R&D dollars come from, and the second is that the ecosystem now includes many different institutions and players, which raises the question of how to drive the greatest return on investment in this new system.

Building New Institutions
Mills asked why institutions are so important to science and whether they are holding it back. Kratsios explained that discovery is increasingly multidisciplinary and no longer maps one to one onto academic fields. Universities are still structured around faculty who work in one discipline within one department. Pointing to the rise of AI as an example of discoveries that require people with different expertise to team up, he said the U.S. needs institutions that match the science of today, such as X-Labs, which are new labs set up to solve a discrete set of problems by bringing multidisciplinary scientists together. Kratsios argued that the current model has let down today's scientists and that the report's reforms are meant to serve them by meeting scientists where they are and helping them make discoveries.

The Role of Universities
Mills noted that the report goes out of its way to say universities have an important role and asked how they fit into a new golden age. Kratsios said researchers working at the frontier of fields like fusion science are often not at universities, which he described as a good sign of institutional diversity. He said universities have struggled to adapt in training young scientists. Most PhD students go into the private sector but are still trained by academics in programs that do not reflect private-sector science. Kratsios pointed to the new Industry-Integrated PhD (I-PhD) program, which matches students with industry so they can finish in four years with their final year at a national lab or in industry, as well as a six-year bachelor's-to-PhD program that lets students graduate at 24 and start making an impact.

Science and Craft
Mills said much of the report focuses on trades and craft, including a section on the marriage of science and craft, and asked why. Kratsios said people forget how important craft is across the full life cycle of a technology in getting to a finished product. He explained that a well-functioning tech ecosystem needs people on assembly lines who are good at building with their hands. He argued that offshoring manufacturing is bad for the country, since the farther manufacturing is from R&D, the bigger the problems become when trying to develop new technology.

Grant Duration and Merit Review
Mills asked whether Kratsios is worried that science is slowing down and what policies would help. Kratsios said the science advocacy establishment has focused on how much money goes into the science machine rather than how the machine operates, which he argued is hurting science. He said it makes no sense that the vast majority of government grants run about two and a half years no matter the project, given the huge variance in how long science takes. He called for more varied grant durations. He added that it takes 18 months to receive funding, which should be quicker. On merit

review, Kratsios said risky ideas are not brought forward because scientists know their peers will not accept them. He pointed to new golden tickets that let each individual member of a review panel unilaterally decide to fund a grant they like, creating opportunities for less conventional ideas to be funded.

Metascience
Mills asked Kratsios to expand on the report's new approach to metascience, which Kratsios described as the science of science. Kratsios said the government should try ideas like golden tickets and run tests to see whether they produce good research. He argued that no one is really looking at how research dollars are spent even as the government keeps spending them.

Reception of the Report
Mills asked about the report's reception and whether its policy framework can work in a polarized environment. Kratsios said the ideas are not partisan and that many of the reforms are long overdue. He acknowledged pushback from the science press establishment but said overall support has been strong.

Working with Established Institutions
Mills asked how the administration strikes a balance with established institutions. Kratsios said concerns about brain drain are not real and that everyone needs to come along and agree. He pointed to 44 universities that have agreed to four-year PhDs with one year spent in industry and to 14 universities that came together to create a Southeast compute hub for the age of AI.

AI Oversight and Safety
During audience questions, an attendee asked which government agencies handle these issues. Kratsios said the National Security Agency (NSA) and the Center for AI Standards and Innovation (CAISI) would work with the results of AI testing as they relate to government. Asked how to balance regulations and guardrails with rapid growth in AI, Kratsios said the administration wants rapid growth but also wants it to happen safely, pointing to the President's launch of the Super Intelligence Force and to the companies that signed the White House Accord on Super Intelligence to make sure the right AI checks are in place.

Startups and Commercialization
An attendee asked what role startups play in this space. Kratsios said there has been no better time to start a company in America than today, and that much of the effort toward a new golden age involves helping startups grow. He explained that science and technology do not follow a linear path, with research often moving back and forth, and said startups are good at figuring out what can be commercialized.

High-Skilled Immigration
Another attendee asked whether industry is wrong about the importance of bringing in high-skilled workers and whether the U.S. should actually be against bringing them here. Kratsios said the U.S. must not allow these programs to be abused in ways that distort the intent Congress put into law. He explained that the administration's H-1B reforms are meant to return the program to what Congress intended.

Science Education
An attendee asked about Kratsios' vision for universities at the bachelor's and master's level. Kratsios said he believes in the broader ecosystem and that the U.S. needs to find ways to bring more Americans into the craft side of science. He also stressed encouraging science from kindergarten through 12th grade. He noted that the administration brought back the NSF Young Scholars program, which was cancelled in the 1990s, to get young people excited about the sciences.

NSF Leadership and Spending
An attendee asked why there is no new NSF director, why the NSB was eliminated, and why NSF has not spent $1 billion that Congress appropriated. Kratsios said President Trump has nominated Jim O'Neill to lead NSF and the administration is waiting on the Senate to confirm him. On the NSB, he pointed to a Supreme Court decision holding that board members who are not Senate confirmed cannot make constitutional decisions. He explained that this is why they were removed and said the administration wants to add more members. He added that NSF appropriations are available over a two-year window by law and that the agency is well on track to spend the funds.

Panel Discussion on Pathways and Obstacles for Science Reform Today.

- Arthur Daemmrich, Director, Arizona State University Consortium for Science, Policy and Outcomes (CSPO)
- Jenn Gustetic, Director of Metascience Policy, Institute for Progress (IFP)
- Asad Ramzanali, Director of Artificial Intelligence and Technology Policy, Vanderbilt Policy Accelerator
- Ian Banks, Director of Science and Innovation, Foundation for American Innovation (FAI) (*Moderator*)

Public Views on Science
Banks asked Daemmrich about his work at CSPO. Daemmrich said CSPO ran three public forums that did not show the public rejecting science, and suggested there is not really an anti-science sentiment. He said participants wanted both the federal government and universities to represent science, and that they felt unrepresented, asking not for a decision-making role but for a voice in deciding what science gets funded. Daemmrich explained that participants did not want to discard merit review but wanted complements to it, such as golden tickets, and said people want a system more connected to public purpose that still respects the old system by preserving expertise and creating more ways for it to adapt.

Trust and Public Purpose
Daemmrich said the U.S. has struggled for a while with a decline in deference to expertise and that institutions need to do more work on how they approach and build trust. He added that outcomes need to improve and that not everything comes down to market economics, using a better metro system as an example of an improvement driven by efficiency rather than market output.

Institutions as Constructive Partners
Banks asked how institutions can be constructive partners in an evolving world of science. Daemmrich said research universities were asked in the past to help launch startups, and that the next stage is for universities to serve as regional technology and economic innovators. He explained that where universities are located matters more now, with more focus on their communities than on a global mission, arguing that universities want to solve world problems but may need to solve problems in their own area first. Gustetic said institutions have different weaknesses depending on the institution, and pointed to the investment system the U.S. has set up as a strength the country already has but needs to lean into and promote.

Stable Funding and the Budget Process
Daemmrich said science funding has been fairly consistent and tied to the state of the economy, and asked Gustetic to clarify the funding model she described, questioning whether it is simply a matter of how the U.S. handles its investments. He asked whether her goal is to remove science funding from the annual process, much like Social Security, and sidestep the professional political debate in which scientific associations take part. Gustetic responded that resource allocation should be a public

discussion, but said agencies like the National Institutes of Health should not be limited to one-year funding and should be able to carry some money over to the next year, since no one can predict when barriers will arise. She pointed to blue-sky research as an area where government compromises are shaped by timing. Another panelist added that he strongly dislikes the annual budget process and that this flexibility should extend to other areas.

The Importance of Metascience
Banks asked Gustetic to describe the importance of metascience to the future of the science enterprise. Gustetic said the U.S. has a remarkably productive R&D market, with every federal dollar invested in R&D returning about $4, but that there are still ways to improve investment in science. She said it is important to study how agencies make decisions while understanding how hard that is when they are in the middle of running those engines, noting that the Foundations for Evidence-Based Policymaking Act (Evidence Act) tried to build this kind of evidence and, despite weaknesses in its design, offers lessons from what it discovered. Gustetic pointed to the Build America Caucus as a group on the Hill that wants to expand metascience reforms, and argued that these reforms must be designed to be sustainable and bipartisan to last, since science takes a long time.

Funding Stability and Independence
Banks asked where people are too optimistic or too pessimistic about institutions. Gustetic said a stable funding base needs to be agreed upon, which would create more efficiency than repeatedly adjusting it, and that the U.S. also needs to find the most talented scientists from around the world. She argued that experimental capacity must be seen as independent, since experimentation cannot be political, and that the resource allocation process should be public. This way, people understand how funding is spent and can hold leaders accountable.

Building a Durable Consensus
Banks asked Ramzanali how to build a politically durable consensus. Ramzanali said laws, especially those that create institutions and move billions of dollars, are far more durable than executive orders and presidential memoranda, and far more durable than visions. He argued that actions mean more than words, saying he does not know many scientists who think the U.S. is headed toward a golden age other than those in AI with stakes in companies, and pointed to the administration freezing funding at universities and calling for roughly 50% cuts at NSF. Ramzanali said experimentation helps put forward new ideas, but asked how many golden tickets have been issued and whether they worked. He described American R&D as the goose that lays the golden egg, and said he fears the U.S. is killing the goose, noting that AI companies are full of former academic PhDs who got their tools in academia and asking how the system can be both broken and leading the world.

AI and the Political Economy
Ramzanali said the report needs to be put in the context of today's political economy, explaining that he wants to promote the AI agenda in the report, but people are very anti-AI right now. He said he is hopeful when he sees AI finding ways to save lives, but dislikes claims that AI can help with biomedical problems when there is no statistical evidence that it is actually saving lives, urging the field to stop focusing on chatbots.

The National Science Board
Ramzanali said he thinks a lot about the firing of the NSB, describing the board as a specific government mechanism. He said the administration justified removing the board's members by pointing to a Supreme Court decision, but that he does not think that was the real reason. Ramzanali argued that even if someone believes the NSB should not exist, the right approach is to have Congress change the law to eliminate or restructure it rather than firing its members, adding that the mechanisms being used feel somewhat made up and that he wishes the administration's goals were clearer.

Attracting Talent to Public Service
Banks asked what it would take to make these science roles more attractive in public service. Gustetic pointed to reforms around hiring, procurement, and grant management, cautioning that these processes could be made either too hard or too easy, both of which would be bad. She said the issue is not just about finding the right people, arguing that fixing the underlying machinery of government matters for everything it does, not only science. Ramzanali said the Office of Personnel Management aims to standardize hiring across a government that employs park rangers, AI specialists, and economists at the Fed, which he called incredibly difficult and not useful, arguing that the U.S. must accept that the government has diverse needs.

Accelerated PhD Programs
An attendee asked for the panel's thoughts on accelerated PhD completion. Gustetic said metascience should be applied to these questions, including tracking whether students stay in science or go into industry after their PhD, so policymakers understand the outputs. Daemmrich pointed to a school in Switzerland where students learned science on older equipment before moving to new computerized tools, noting that even students not on that career track learned these skills, and said that in America the craft side of science is not seen as a viable option. Ramzanali added that science and technology jobs are not just for PhDs.

Commercializing Federally Funded Research
Another attendee asked about the biggest bottlenecks to commercializing federally funded technology and who needs to act. Daemmrich said the U.S. has turned federal research labs and universities into an intellectual property pricing scheme, noting that they produce about 20,000 patents a year but that a patent is not a product. He said people used to spend time in both industry labs and universities and that the U.S. needs that dual, collaborative model again. Gustetic said institutions would need to develop dual approaches, asked whether scientists and technologists can be trained to be better businesspeople, and noted that even if AI generates many ideas, people still need to build them and take charge of the work.

The Role of National Labs
An attendee asked about the role of national labs in that process. Gustetic said national labs house important intramural talent and should be better optimized for these roles, cautioning that labs sometimes end up doing far more than their core mission. Daemmrich said the U.S. needs to find a way to do this work within the national labs or build facilities adjacent to them.